School of Engineering and Architecture

Department of Electrical and Electronics Engineering

IOT-BASED AIR QUALITY MONITORING SYSTEM

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A project submitted in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Electrical and Electronics Engineering in the Department of Electrical and Electronics Engineering at Meru University of Science and Technology.

April, 2026

DECLARATION

We hereby declare that this project proposal is our original work except as cited in the references and has not been presented for the award of a degree in any other University.

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LIST OF ABBREVIATIONS

ESP32 - Espressif 32-bit Microcontroller

IoT – Internet of Things

DHT11 - Digital Humidity and Temperature Sensor

LPG - Liquefied Petroleum Gas

MQ - Metal Oxide Semiconductor Gas Sensor Series

MQ7 - Carbon Monoxide Gas Sensor

PM - Particulate Matter

$PM_{2.5}$ – Particulate Matter with diameter $\leq$ 2.5 micrometers

PM_{10}- Particulate Matter with diameter $\leq$ 10 micrometers

ML - Machine Learning

CO - Carbon Monoxide

RMSE - Root Mean Square Error

MAE - Mean Absolute Error

RF – Random Forest

IS – Isolation Forest

DT – Decision Tree

ABSTRACT

Air quality degradation and hazardous gas emissions pose significant challenges to public health, safety, and environmental sustainability. Urbanization and industrialization have increased the release of pollutants such as carbon oxides, methane, and particulate matter (PM2.5 and PM10), while domestic environments remain at risk from leaks of liquefied petroleum gas (LPG) and other combustible gases. These pollutants are linked to respiratory illnesses, cardiovascular diseases, climate change, and fire hazards. Existing monitoring systems are often centralized, expensive, and inaccessible for real-time decision-making at the household or community level. This project proposes an IoT-based air quality monitoring system capable of real-time data acquisition, cloud-based storage, and predictive analytics. The system integrates low-cost MQ gas sensors, particulate matter sensors, and environmental sensors with an ESP32 microcontroller for data collection and processing. Data is transmitted to Firebase for secure storage, visualization, and alert dissemination through a web application. Predictive models are employed to forecast pollution levels, detect anomalies, and assess potential fire hazards, while an actuation and safety response mechanism ensures timely mitigation. The proposed system provides a scalable, resource-efficient solution for continuous air quality monitoring, early warning, and risk management across urban, domestic, agricultural, and healthcare environments.

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CHAPTER 1: INTRODUCTION

1.1 BACKGROUND OF STUDY

Air quality degradation has become a pressing environmental and public health concern globally. Rapid urbanization, increased industrial activities, transportation, and population growth have significantly elevated the release of harmful pollutants such as particulate matter ($PM_{2.5}$ and PM_{10}), methane, carbon oxides, and volatile organic and inorganic compounds [1]. These pollutants are directly associated with respiratory illnesses, cardiovascular diseases, allergies, and climate impacts [1]. In addition to outdoor pollution, domestic environments face risks of hazardous gas leaks such as liquefied petroleum gas (LPG) and methane, which can lead to poisoning, fire outbreaks, and explosions [2].

Although government-operated monitoring stations exist in major cities, current air quality monitoring approaches remain centralized, expensive, and inaccessible to individuals at the household level [1]. This creates a gap in real-time localized monitoring that can help mitigate risks [3].

Recent advancements in Internet of Things (IoT), cloud computing, and machine learning present a promising opportunity to develop affordable and scalable monitoring systems [3]. The proposed IoT-Based Air Quality Monitoring System integrates low-cost sensors with cloud-based storage and real-time data visualization to deliver accessible environmental insights [4]; [5]. By leveraging MQ-series gas sensors, particulate matter sensors, temperature and humidity sensors (DHT11), and ESP32 microcontrollers, the system offers continuous surveillance of air pollutants and hazardous gases [4]; [5]. The integration of predictive analytics further enhances the capability to detect anomalies and forecast pollution trends [3]; [6]; [7].

The integration of an actuation and safety response system strengthens hazard mitigation at the household level [2]. Environmental factors such as temperature and humidity also influence sensor readings and system accuracy, making real-time calibration and compensation necessary [8]; [9]. The system, therefore, represents a timely initiative toward improving environmental health, public safety, and awareness by bridging the gap between centralized air quality monitoring and household-level accessibility [5].

1.2 PROBLEM STATEMENT

Air pollution and hazardous gas exposure remain persistent problems in urban and domestic environments due to increasing industrial activity, vehicle emissions, and the widespread use of gas-powered appliances. In many residential areas and small business premises, exposure to pollutants such as carbon monoxide and combustible gases often goes undetected until health symptoms, fires, or explosions occur, significantly increasing the risk of injury, property damage, and loss of life. Existing air quality and gas monitoring systems are largely centralized, expensive, and designed for regulatory or industrial applications rather than individual households or small enterprises. As a result, they are inaccessible to most users and fail to provide continuous, real-time, and location-specific information at the point of exposure, leaving individuals without timely awareness or early warning of deteriorating air quality or gas leakages in their immediate surroundings. The absence of a resource-efficient, real-time, and localized monitoring solution therefore creates a critical safety and public health gap, causing communities to remain reactive rather than preventive in managing air pollution and gas-related hazards. Addressing this gap requires an economical, scalable, and accessible monitoring system capable of detecting air pollutants and hazardous gases in real time, generating timely alerts, and enabling preventive response through automated actuation, which is the aim of the proposed IoT-Based Air Quality Monitoring System.

1.3 OBJECTIVES

Main objective

To design a scalable IoT-Based Air Quality Monitoring System with real-time detection, prediction, alerts, and actuation.

Specific objectives:

a. To measure and log air pollutants (PM2.5, PM10, CO, LPG) along with temperature and humidity using low-cost sensors connected to an ESP32 microcontroller.
b. To transmit collected data to a cloud platform for real-time visualization, predictive modeling of air quality trends, and threshold-based alert generation through a web interface.
c. To implement automated actuation mechanisms that respond to detected hazards or predicted unsafe conditions, such as activating ventilation or triggering safety alerts.

1.4 JUSTIFICATION

Air quality monitoring solutions are predominantly confined to government-operated or industrial-scale installations, which limits accessibility for households and small businesses. Consequently, individuals are frequently exposed to invisible yet hazardous pollutants and gas leaks without timely detection or warning. An economical and scalable IoT-based Air Quality Monitoring System addresses this gap by enabling early detection and automated alerts that enhance public safety. It provides accessible, real-time environmental data that supports informed decision-making at the individual and community levels, thereby reducing exposure-related health risks. By promoting community awareness and integrating IoT infrastructure with machine learning analytics, the system enables continuous monitoring and early detection of air quality issues. It provides a practical, research-based solution for improving environmental health and public safety.

1.5 SCOPE

This project focuses on designing and implementing an IoT-based air quality monitoring system using low-cost gas sensors, particulate matter sensors, environmental sensors, actuators, and ESP32 microcontrollers. The scope includes real-time data collection, wireless transmission to a cloud platform (Firebase), web-based data visualization, safety response, and predictive modeling for anomaly detection and forecasting. The project will cover; hardware integration, firmware development, cloud connectivity, web application interface, alert systems, and evaluation of system performance. This project, however, does not cover nationwide monitoring or industrial certification standards. Instead, the focus remains on household, urban, agricultural, and healthcare environments where feasible and cost-effective deployment is needed.

CHAPTER 2: LITERATURE REVIEW

Air quality monitoring has become increasingly important due to rising pollution levels and associated health risks. Traditional reference-grade monitoring stations provide high measurement accuracy but are expensive and sparsely distributed, limiting their spatial coverage and accessibility. Bagkis et al. [1]; Kumar et al. [2] highlighted the technological evolution of low-cost sensor networks and emphasized the need for scalable, distributed monitoring solutions. Similarly, Garcia et al. [3] noted that centralized systems restrict accessibility and integration with modern digital infrastructures.

Recent research has therefore shifted toward low-cost sensors and IoT-based systems to improve real-time environmental monitoring. Concas et al. [4] surveyed low-cost outdoor air quality monitoring systems, discussing calibration challenges, sensor drift, and cross-sensitivity effects. Their work emphasized proper calibration strategies for reliable deployment at scale.

2.1 Technologies Used in Air Quality Monitoring

Low-cost sensors are increasingly preferred for distributed monitoring due to their affordability and compact size. Concas et al. [4] explained that sensor drift and environmental interference can affect measurement accuracy over time. Environmental conditions further impact sensor performance. Spinelle et al. [14] showed that temperature and humidity variations influence sensor sensitivity and stability, while Casari and Po [8] reported that high humidity inflates particulate matter readings due to hygroscopic growth.

To address these challenges, IoT architectures and machine learning techniques are widely used. Zaid et al. [10] developed an IoT-based low-cost sensor network capable of reliable wireless data transmission, real-time monitoring, and cloud storage. Ramesh et al. [11] reported that integrating sensors with microcontrollers and cloud dashboards enables real-time monitoring and early hazard detection. Garcia et al. [3] also highlighted the use of AI to improve predictive analytics and system automation.

Machine learning has improved sensor reliability and data accuracy. Tastan [6] applied machine learning for sensor calibration and reported improvements in measurement accuracy and stability. Kim and Seung-Hyun [7] reviewed a decade of machine learning methods for correcting sensor drift and noise, including Random Forest, SVM, and deep learning. Idir et al. [12] compared predictive models for urban air quality using mobile and fixed sensors, showing that data fusion enhances spatial resolution and forecasting accuracy.

2.2 Review of Existing Systems

Several studies have demonstrated the practical deployment of low-cost monitoring systems. Gatari et al. [13] conducted one of the earliest studies in Kenya using calibrated particulate matter sensors, validating their accuracy against reference instruments. Njeru et al. [14] deployed a PM2.5 sensor network in Mombasa, Kenya, reporting frequent exceedances of health-based air quality guidelines, indicating high exposure risks.

Community-integrated approaches improve monitoring effectiveness. Manshur et al. [15] implemented citizen-science air quality monitoring in a Kenyan informal settlement, showing both technical feasibility and social value. Bowyer et al. [16] demonstrated that participatory arts-based methods, such as theatre, music, murals, and school outreach, increase awareness and participation in air pollution studies.

Indoor and household-level monitoring systems are increasingly emphasized. Othman et al. [5] reviewed low-cost IoT-based indoor air quality monitoring systems, highlighting low power requirements and suitability for household deployment. Kumar et al. [2] developed a gas leakage detection system with automated alerts, showing the importance of safety response mechanisms in household monitoring.

Ramesh et al. [11] identified safety response mechanisms as a key component of air quality monitoring systems, enabling real-time hazard detection and mitigation.

2.3 Research Gaps

Despite advances in low-cost sensors, IoT platforms, and machine learning, several gaps remain. Household-level access to reliable air quality monitoring is still limited. Many current systems lack real-time alert mechanisms and predictive analytics integration. Community engagement strategies are often underutilized, reducing participation and trust. Finally, sensor performance can be affected by environmental conditions, and more work is needed to maintain accuracy under varying temperature and humidity. These gaps justify the development of an integrated solution combining distributed sensors, IoT communication, cloud-based data management, machine learning analytics, and community engagement to provide accessible and actionable air quality information.

CHAPTER 3: METHODOLOGY

3.1 SYSTEM REQUIREMENTS

1. ESP32 microcontroller.
2. DHT11 sensor - Temperature and humidity.
3. Mq7 sensor - carbon monoxide(CO).
4. MQ8 sensor – Hydrogen Gas
5. MQ5 sensor - LPG
6. Nova PM sensor Particulate matter.
7. Breadboard and stripboard.
8. 12 V battery and battery charger.
9. Powerbank.
10. Buzzer.
11. Relay.
12. Jumper wires.
13. Firebase
14. Arduino IDE
15. Visual studio code
16. GitHub
17. LCD Display.
18. Voltage regulator

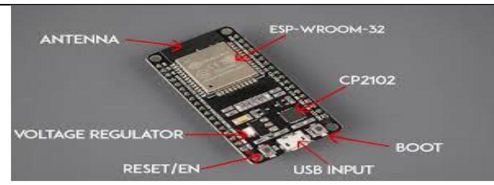
<table>
<tr>
<td>
Figure 1:Passive Buzzer</td>
<td>

Figure 2: ESP32 Board</td>
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Figure 3: DHT11 Sensor</td>
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Figure 4: LCD Display</td>
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Figure 5: PM nova Sensor</td>
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Figure 6: MQ8 Sensor</td>
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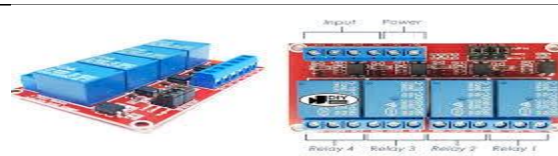
Figure 7:MQ5 Sensor</td>
<td>
Figure 8: 4_way relay Channel</td>
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Figure 9: MQ7 Sensor</td>
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Figure 10: Voltage Regulator</td>
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3.2 SYSTEM DEVELOPMENT

3.2.1 Circuit Development

The sensors' Vcc pin was connected to the positive terminal (+5V) of the power supply unit (PSU) and the ground pins was connected to the ground terminal (negative terminal) of the power supply unit. The sensors could not be supplied by the 5V pin of the ESP32 because the total power consumption of the sensors exceeded what could be supplied by the ESP32's 5V pin and doing so would have overloaded the onboard voltage regulator that would have damaged the ESP32 board. The ground pin of the ESP32 and the power supply unit's ground terminal were connected together to provide a common ground for the sensors ensuring that the signal values from the sensors were stable.

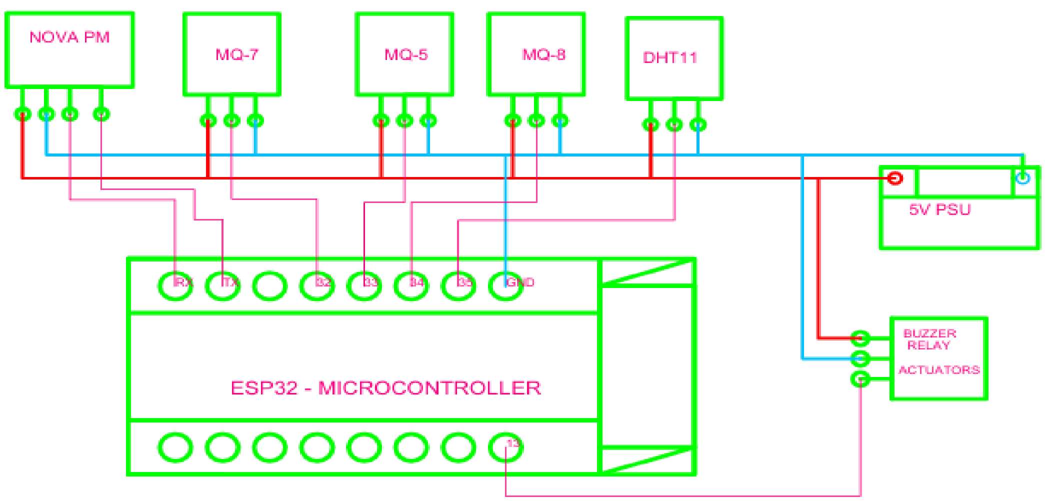


Figure 11: Circuit Connection

3.2.2 Database and Web Development

The real-time database was developed on firebase studio for storage of the sensor values transmitted from the ESP32 node. The web scripts were done on visual studio code and configuration keys for the database to enable seamless communication between the database and the web page was inserted accordingly. The web app was tested before deploying to the GitHub pages for hosting.

3.2.3 Firmware Development

The program that enabled the ESP32 microcontroller to read values from the sensors, connect to the internet, perform necessary signal processing and package the data for transmission to the firebase database was developed in Arduino IDE. The configuration keys for the database were embedded into the program to enable seamless communication between the ESP32 and the firebase real-time database. This program aided in data collection.

3.3 DATA COLLECTION

DHT11 sensor was used for temperature and humidity monitoring, PM nova sensor was used for particulate matter ($PM_{2.5}$ and PM_{10}) monitoring, MQ5 was be used for LPG monitoring, MQ8 was used for hydrogen monitoring, and MQ7 was used for carbon monoxide (CO) monitoring.

The sensors took readings of the respective air constituents and transmitted the values in terms of a voltage signal to the analog pins of the ESP32 board, the signal was then digitized inside the ESP32 using the analog to digital converters (ADCs) inside the ESP32 board. The values were quantized within a 12-bit range of 0-4095 depending on the strength of the signal received and transmitted via internet to the firebase real-time database for storage. Through Hypertext Transfer Protocol (HTTP), the data was conveyed to the web application for real-time visualization.

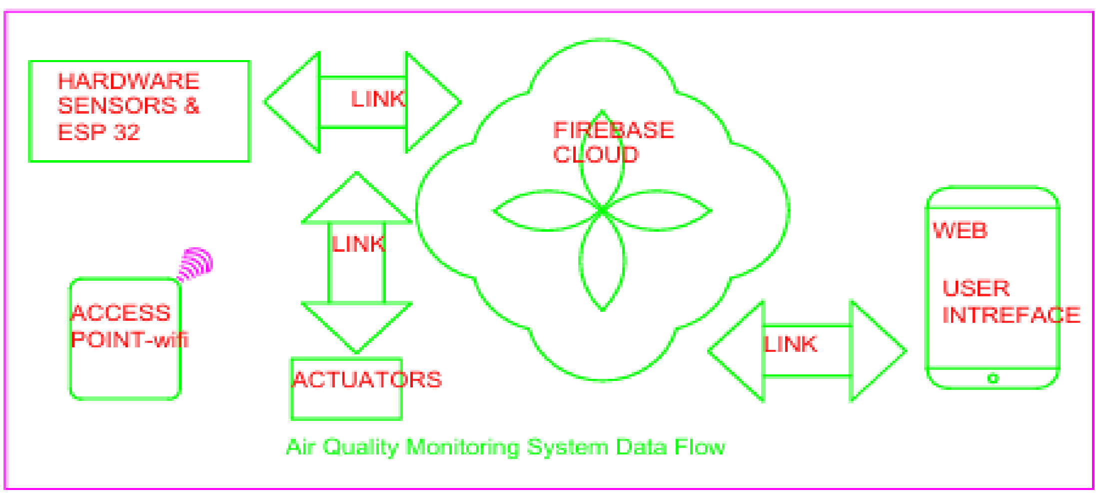


Figure 12: Flow of Data

The data collection exercise spanned for 27 days and all of it was stored in the firebase database where later it was later accessed for analysis. The transfer of the data to the firebase database occurred after every 30 seconds. This reduced the number of requests and responses (HTTP protocol) that the ESP32 has to make per unit time during transmission of the data to the cloud based database and also reduced network overhead and improved system stability.

SYSTEM OPERATION

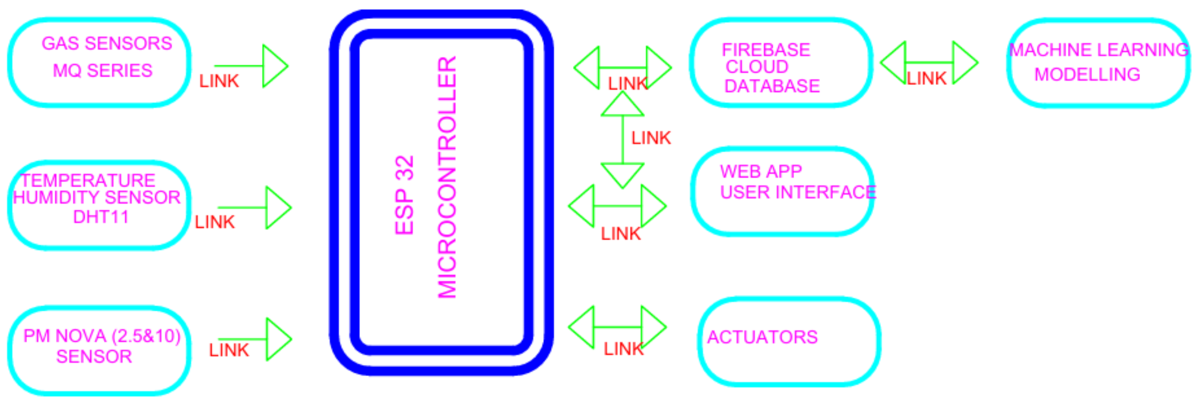


Figure 13: System Block Diagram

Data collected by the sensors was passed to the ESP32 microcontroller for calibration, packaging and transmission to firebase for the purposes of storage. From firebase the data was accessed by the web application for real-time visualization. The machine learning unit that was responsible for short term prediction and anomaly detection accessed the data from firebase and performed the necessary analytics. Depending on the results of the analytics alerts were disseminated accordingly. The actuator was activated or deactivated necessarily depending on the output of the analytics.

3.4 DATA ANALYSIS METHODS

Sensor data was collected and stored in the Firebase database at a fixed interval of 30 seconds for a duration of 27 days. The total number of data points was estimated using the expression:

$$(Days \times Hours\_In\_A\_day \times Minutes\_In\_Hour \times Seconds\_In\_minute) \div 30$$

For a 27-days data collection period, this results in 66,919 data points per sensor**.** The collected dataset was preprocessed to remove missing values, handle outliers, and ensure timestamp consistency before further analysis. The cleaned data was used to train simple machine learning models aimed at short-term prediction and anomaly detection.

3.4.1 Machine Learning Modeling

The following machine learning models were employed in this project:

I. Decision Tree (DT)

Decision Trees are supervised learning algorithms used for both classification and regression tasks. They operate by recursively splitting the input data based on feature values, forming a tree structure where internal nodes represent decision rules and leaf nodes represent output predictions. Decision Trees are easy to interpret, computationally efficient, and suitable for structured sensor data. However, they are sensitive to small variations in the dataset and are prone to overfitting. To reduce this effect, tree depth and splitting criteria will be controlled during training.

II. Random Forest (RF)

Random Forest is an ensemble learning technique that combines multiple Decision Trees to improve prediction accuracy and robustness. Each tree was trained on a random subset of the dataset, and final predictions were obtained through averaging (for regression). Random Forest models were less sensitive to noise and provide improved generalization compared to individual Decision Trees. Their main limitation was increased computational cost and reduced interpretability due to the ensemble structure.

III. Isolation Forest (IF)

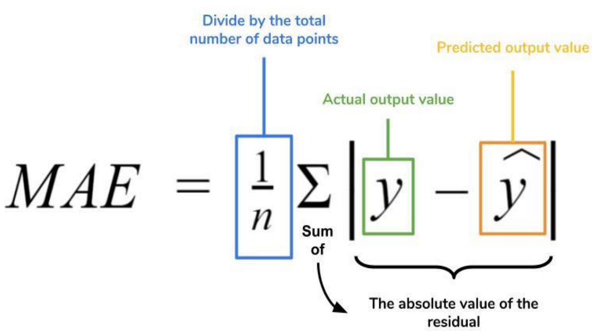
Isolation Forest is an unsupervised learning algorithm designed for anomaly detection. It works by isolating data points through random partitioning, where anomalies require fewer splits to be isolated compared to normal data points. This model was used to detect abnormal pollution patterns such as sudden gas concentration spikes that deviated from normal operating conditions.

3.4.2 Models' Training and Deployment

The collected data was subdivided in the ratio 80:20 where 80% of the data was used for the models' training and the remaining 20% was used for testing the models. The models' training was done locally on the PC. The predicted values (random Forest, RD) were displayed on the web dashboard and incase of anomaly detection (Isolation forest, IF) alerts were triggered on the web dashboard for action.

3.4.3 Model Evaluation

a) Mean Absolute Error (MAE) - MAE measured the average magnitude of errors between the predicted values and the actual sensor readings, ignoring the direction of the errors. It was calculated as the arithmetic mean of the absolute differences between predicted and actual values.



b) Root Mean Square Error (RMSE) - RMSE measured the square root of the average squared differences between predicted and actual values:

$$RMSE = \sqrt{\sum_{i=1}^{n} \frac{(\hat{y}_i - y_i)^2}{n}}$$

$\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_n$ are predicted values

$y_1, y_2, \ldots, y_n$ are observed values

n is the number of observations

Unlike MAE, RMSE penalized larger errors more heavily, making it more sensitive to outliers. Together with MAE, RMSE provided a comprehensive view of the model's prediction accuracy.

3.4.4 SYSTEM FLOWCHART

The flowchart shows flow of the system's logic. Data is collected by sensors and ESP handles necessary processing for transmission to firebase, from firebase date is visualized on the web app, the machine learning unit (Python server) handles the short term prediction and anomaly detection and from the results alerts are disseminated and necessary actuation mechanisms activated.

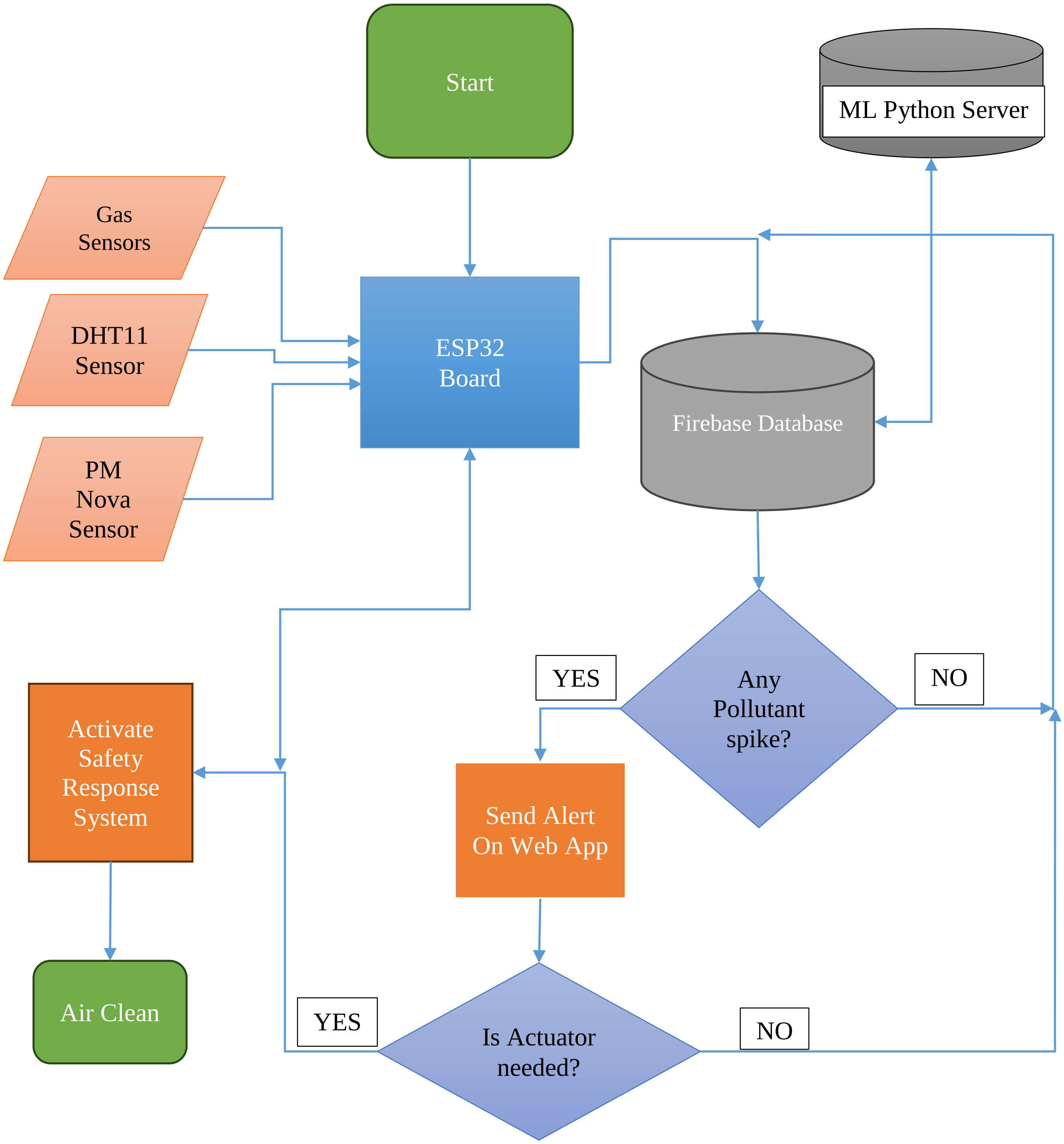


Figure 14: System Flow Chart

3.5 PROTOTYPE TESTING

The developed prototype was tested under both controlled and real-world operating conditions to verify system functionality and performance. Controlled testing was carried out using simulated pollutant sources to evaluate sensor response, threshold detection, and alert triggering behavior. Real-world testing was conducted in a domestic environment to assess system reliability under normal environmental variations such as temperature changes, ventilation differences, and background pollution levels. Key performance parameters evaluated include sensor responsiveness, alert latency, cloud data update consistency, and web dashboard functionality. The results of these tests were used to validate that the system met its intended functional and operational requirements.

3.6 ETHICAL AND SAFETY CONSIDERATIONS

1. Gas sources used during controlled testing were handled in well-ventilated environments to minimize the risk of accidental exposure.
2. Testing procedures were designed to ensure that no individuals are exposed to hazardous gas concentrations at any stage of system evaluation.
3. The system collects only environmental sensor data for research and performance evaluation purposes. No personal or sensitive user data was recorded, and access to stored data was restricted to authorized users.

CHAPTER 4: RESULTS

$R^2 = 1 \rightarrow$ perfect prediction

$R^2 = 0 \rightarrow$ model performs like mean baseline

$R^2 < 0 \rightarrow$ model performs worse than mean prediction

4.1 DECISION TREE MODEL

Data splitting strategy

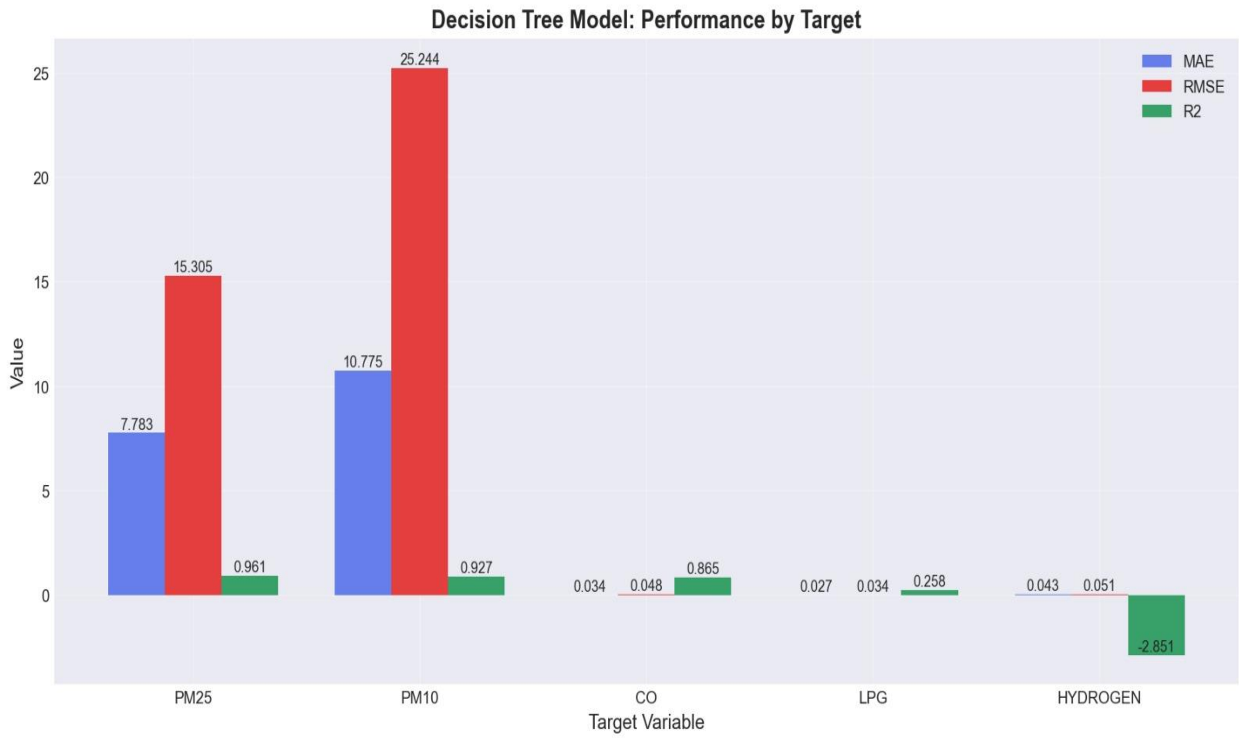
The dataset was divided into training and testing sets using an 80/20 split ratio.

Total dataset size = 66,919 records

Training set (80%): 66,919 × 0.8 = 53,535 records (approximately)

Testing set (20%): 66,919 × 0.2 = 13,384 records (approximately)

Results



MAE is relatively high, indicating noticeable prediction deviation

RMSE is higher than MAE, suggesting presence of larger errors in some predictions

R^2 is low (approximately 0.31), indicating weak explanatory power

Key Observations:

The model captured only basic nonlinear relationships. It struggled with highly dynamic sensor variations.

It was sensitive to noise and outliers in the dataset.

Limitations

The Decision Tree model had several limitations in this application:

- High tendency to overfit if not properly constrained
- Poor generalization on complex or noisy sensor data
- No inherent capability to model time-series dependencies
- Unstable predictions when small data variations occur

Due to these limitations, it is was not suitable as a final production forecasting model.

Conclusion

The Decision Tree regression model provided a simple and interpretable method for predicting air quality parameters from IoT sensor data. It was effective as a baseline model and useful for initial system validation. However due to its limitations it is less suitable for high-accuracy forecasting tasks. It is therefore primarily used as a reference model for evaluating improvements achieved by ensemble methods (Random Forest, XGBoost) and deep learning approaches (LSTM).

4.2 RANDOM FOREST MODEL

Data spitting strategy

The dataset was divided into training and testing sets using an 80/20 split ratio.

Total dataset size = 66,919 records

Training set (80%): 66,919 × 0.8 = 53,535 records (approximately)

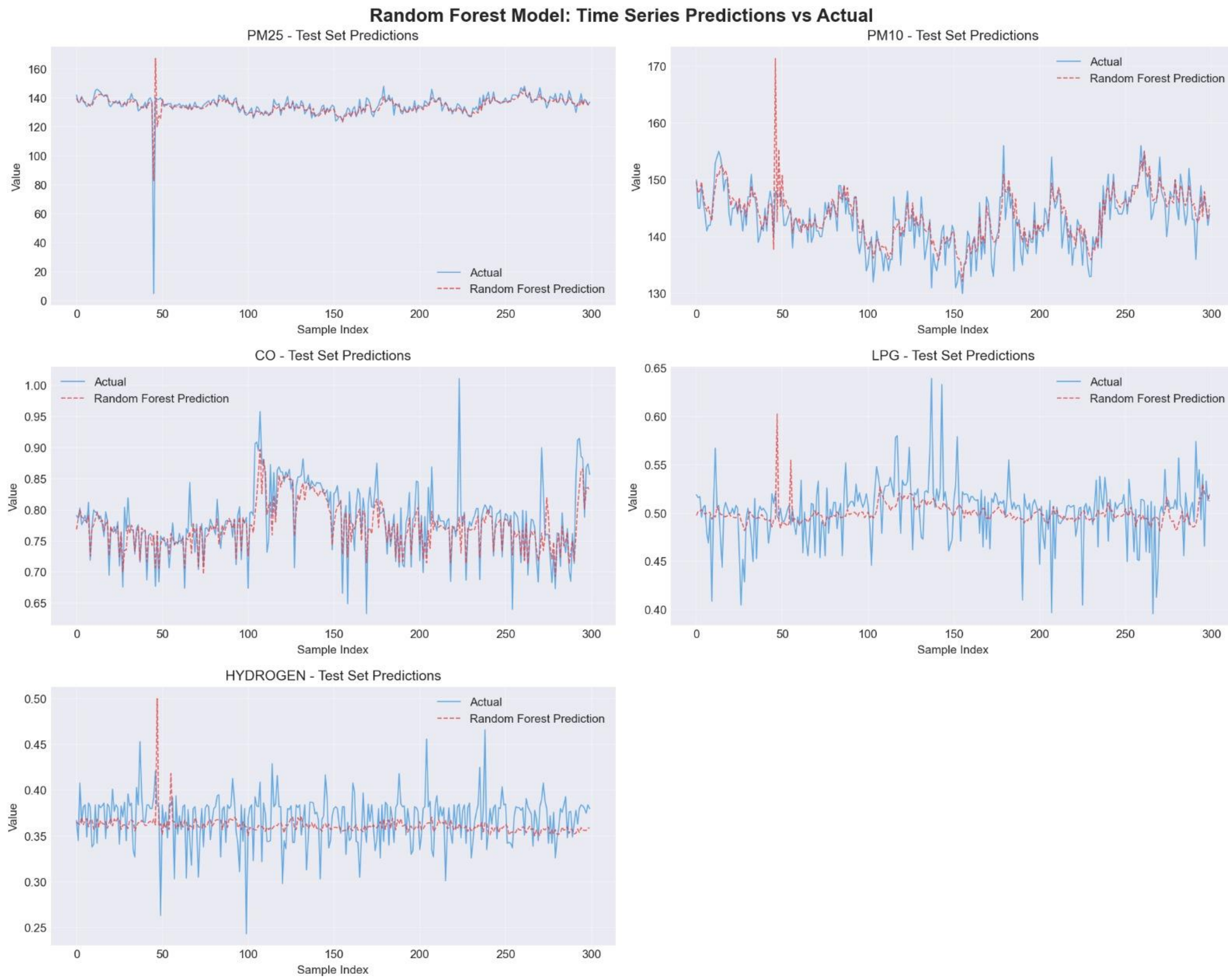
Testing set (20%): 66,919 × 0.2 = 13,384 records (approximately)

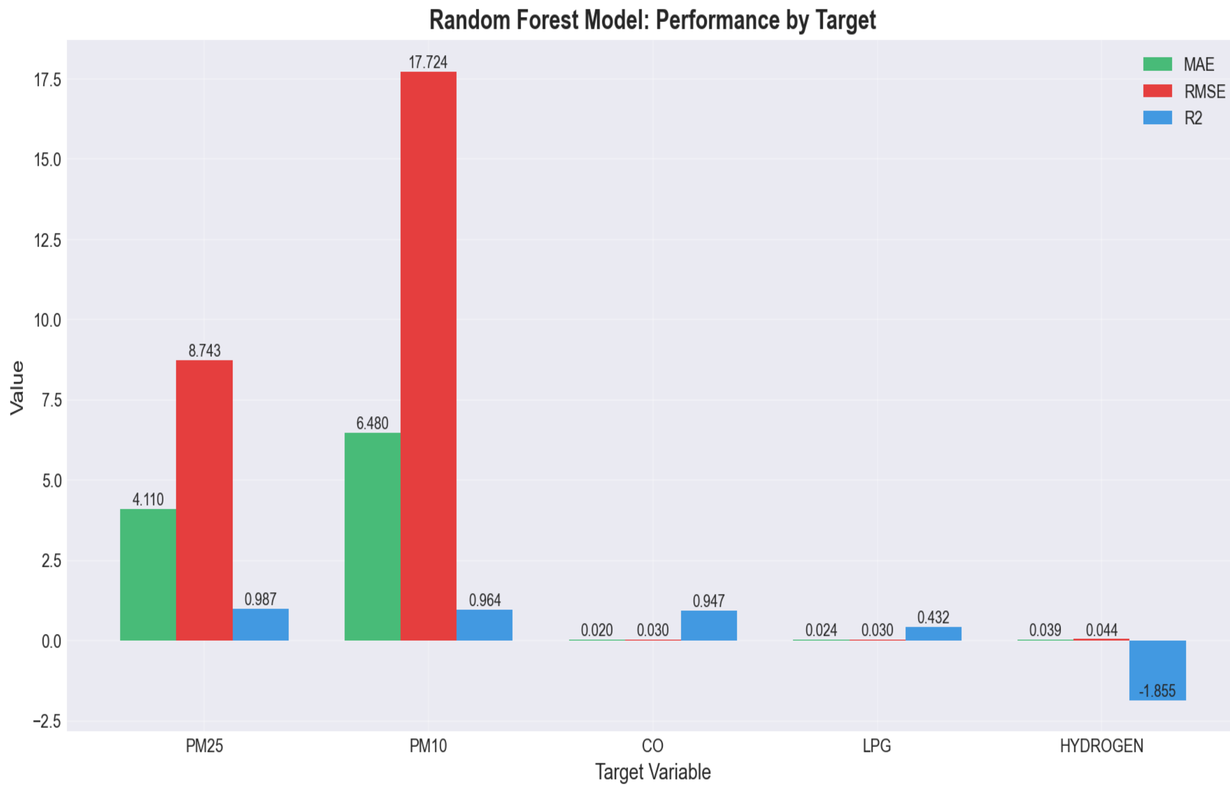
Observed performance:

MAE: =58.04

RMSE: =100.82

R^2: = 0.47





Key Observations:

- Better performance than Decision Tree
- Reduced overfitting due to ensemble averaging
- More stable predictions across noisy sensor data
- Still limited in capturing temporal dependencies

Limitations

Cannot model time-series dependencies

Higher computational cost than single decision tree

Less interpretable due to ensemble complexity

Performance depends on feature quality and tuning

Conclusion

Random Forest improves prediction accuracy and stability compared to a single decision tree by using ensemble learning. It reduces variance and improves generalization on noisy IoT sensor data. However, it still lacks temporal awareness and is not optimal for forecasting future air quality trends. It is best suited as a strong intermediate model between simple regression methods and advanced boosting or deep learning models.

4.3 ISOLATION FOREST

Data Splitting Strategy

Isolation Forest does not use a traditional supervised train-test split. Instead the model is trained on the entire dataset (66,919 records). A contamination parameter defines expected anomaly ratio. Evaluation is performed using threshold-based anomaly scoring.

Evaluation Metrics

Since this is an unsupervised model, evaluation was not based on traditional regression metrics. Instead, evaluation used:

1. Anomaly Score Distribution

   Measured how data points are distributed between normal and abnormal regions

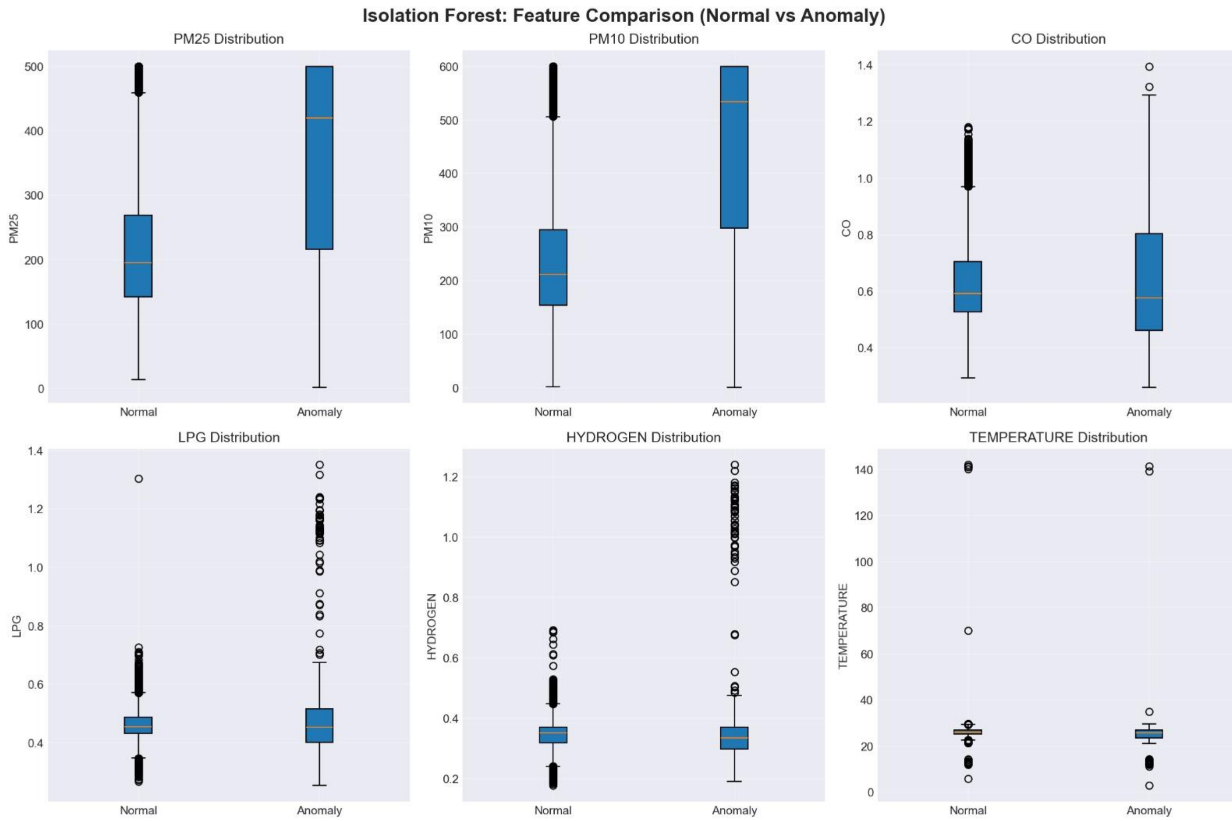
2. Threshold-Based Validation

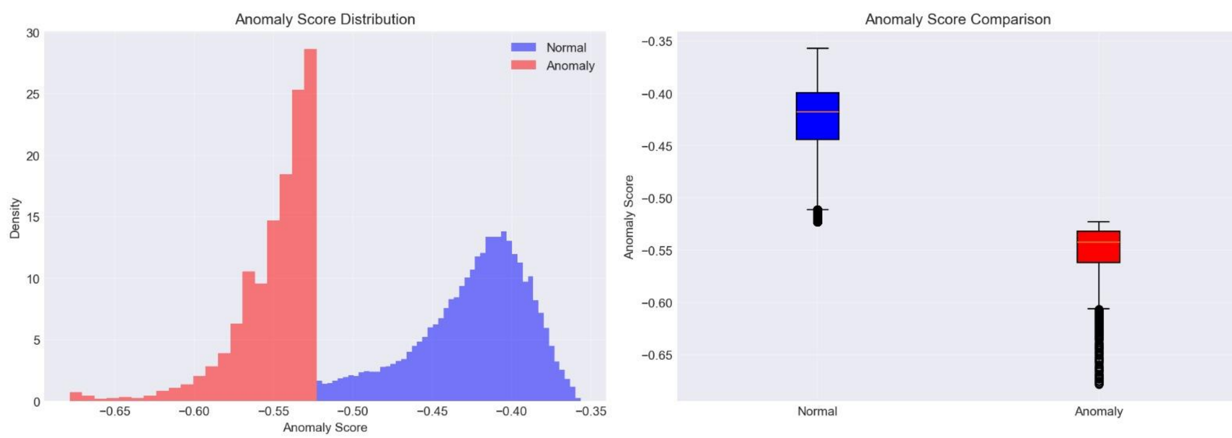
   Domain thresholds were applied (e.g., MQ7 high values) and Model outputs were compared against known environmental rules.

3. Detection Consistency

   Stability of anomaly detection across time-series data

Interpretation of Results





The model successfully identifies extreme deviations in sensor readings.

High MQ sensor spikes are consistently flagged as anomalies

Performance depends heavily on contamination parameter selection

Limitations

Does not predict future values Sensitive to parameter tuning (contamination rate)

Can misclassify rare but valid environmental conditions

No temporal awareness of sensor data

Conclusion

Isolation Forest provides an efficient and scalable method for detecting abnormal air quality conditions in IoT sensor data. It plays a critical role in identifying unexpected environmental events and supporting real-time alert systems. However, it is not suitable for forecasting and must be combined with predictive models for a complete system.

4.4 CONCLUSION

This study successfully achieved its main objective of designing and evaluating a scalable IoT-Based Air Quality Monitoring System capable of real-time air quality detection, monitoring, predictive analysis, alert generation, and safety actuation. The developed system effectively integrated ESP32 microcontroller technology with low-cost sensors (MQ-series, PM, DHT11), Firebase cloud infrastructure, web-based visualization, and machine learning models to create a functional and resource-efficient monitoring framework.

The specific objectives were substantially achieved. First, the system successfully measured and logged key environmental parameters including PM2.5, PM10, carbon monoxide, LPG, temperature, and humidity using integrated sensors. Second, cloud transmission and storage through Firebase, combined with web dashboard visualization, demonstrated that real-time environmental monitoring and threshold-based alerting were effectively implemented. Third, predictive analytics and anomaly detection were successfully incorporated through Decision Tree, Random Forest, and Isolation Forest models, with Random Forest outperforming Decision Tree in prediction accuracy and stability, while Isolation Forest effectively detected abnormal pollution spikes for hazard identification. Although some limitations remain in temporal forecasting due to the absence of advanced time-series models such as LSTM, the system achieved its intended predictive and anomaly detection goals at a practical prototype level.

The identified research gaps were also significantly addressed. The project responded to the lack of affordable household-level air quality monitoring by providing a low-cost, localized, and scalable solution. It bridged the accessibility gap created by expensive centralized systems through IoT deployment, addressed the absence of real-time hazard alerts through automated web-based notifications and safety response mechanisms, and incorporated machine learning to improve system intelligence beyond conventional monitoring systems. Furthermore, by integrating environmental monitoring with predictive analytics and actuation, the study advanced beyond many existing low-cost monitoring systems that primarily focus only on passive observation.

Overall, the research demonstrates that the proposed IoT-Based Air Quality Monitoring System is a viable solution for improving environmental safety, public health awareness, and early hazard prevention in domestic, urban, agricultural, and healthcare settings. While future improvements may include advanced time-series forecasting models, sensor calibration optimization, and broader deployment, the project successfully met its intended objectives and made a meaningful contribution toward addressing the identified technological and practical gaps in accessible air quality monitoring.

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